

Task 1 – Install GPG and Generate a Key Pair Ubuntu/Kali Linux

Install GPG:

```
sudo apt update  
sudo apt install gnupg -y
```

Check installation:

```
gpg --version
```

Generate a key:

```
gpg --full-generate-key
```

Select the recommended options.

Enter:

Real name: Nishit Panchal

Email: your-email@example.com

Use your actual email address when performing the practical.

Check the key:

```
gpg --list-keys
```

Check secret keys:

```
gpg --list-secret-keys
```

Task 2 – Encrypt secret_message.txt

Create the file:

```
nano secret_message.txt
```

Example:

This is my confidential cybersecurity practical message.

Save it.

Find your key:

```
gpg --list-keys
```

Encrypt the file:

```
gpg --encrypt --recipient "your-email@example.com" secret_message.txt
```

This creates:

```
secret_message.txt.gpg
```

The encrypted file cannot be read as normal text.

Task 3 – Decrypt the File

Run:

```
gpg --decrypt secret_message.txt.gpg > decrypted_message.txt
```

View the decrypted file:

```
cat decrypted_message.txt
```

Expected:

This is my confidential cybersecurity practical message.

The original message has been successfully restored.

Task 4 – Simulate Sending an Encrypted File

Export Your Public Key

```
gpg --armor --export "your-email@example.com" > nishit_public_key.asc
```

The file:

nishit_public_key.asc

contains your public key.

The public key can be shared.

Never share your private key.

Import a Classmate's Public Key

If your classmate gives you:

classmate_public_key.asc

import it:

```
gpg --import classmate_public_key.asc
```

Check the imported key:

```
gpg --list-keys
```

Create playlist.txt

```
nano playlist.txt
```

Example:

Top 5 Spotify Songs

1. Blinding Lights
2. Perfect
3. Believer
4. Shape of You
5. Night Changes

Encrypt it using the classmate's public key:

```
gpg --encrypt --recipient "classmate-email@example.com" playlist.txt
```

This produces:

playlist.txt.gpg

Only someone possessing the corresponding private key can decrypt the encrypted file.

Task 5 – WhatsApp/Signal vs GPG

WhatsApp and Signal use end-to-end encryption so that the communicating endpoints can decrypt message content while intermediaries cannot normally read it. Modern messaging systems establish and manage cryptographic keys automatically, whereas GPG normally requires users to generate, exchange, and manage keys themselves. GPG file encryption can be performed using a recipient's public key so that only the corresponding private-key holder can decrypt the file. Therefore, both approaches use public-key cryptography and symmetric encryption concepts, but messaging applications automate key management and provide additional protocols for ongoing conversations.